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ORIGINAL ARTICLE

Metagenomic mining of Indian river confluence reveal functional microbial community with lignocelluloytic potential

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Abstract

Microbial carbohydrate-active enzymes (CAZyme) can be harnessed for valorization of Lignocellulosic biomass (LCB) to value-added chemicals/products. The two Indian Rivers Ganges and the Yamuna having different origins and flow, face accumulation of carbon-rich substrates due to the discharge of wastewater from adjoining paper and pulp industries, which could potentially contribute to the natural enrichment of LCB utilizing genes, especially at their confluence. We analyzed CAZyme diversity in metagenomic datasets across the sacred confluence of the Rivers Ganges and Yamuna. Functional annotation using CAZyme database identified a total of 77,815 putative genes with functional domains involved in the catalysis of carbohydrate degradation or synthesis of glycosidic bonds. The metagenomic analysis detected ~ 41% CAZymes catalyzing the hydrolysis of lignocellulosic biomass polymers- cellulose, hemicellulose, lignin, and pectin. The Beta diversity analysis suggested higher CAZyme diversity at downstream region of the river confluence, which could be useful niche for culture-based studies. Taxonomic origin for CAZymes revealed the predominance of bacteria (97%), followed by archaea (1.67%), Eukaryota (0.63%), and viruses (0.7%). Metagenome guided CAZyme diversity of the microflora spanning across the confluence of Ganges-Yamuna River, could be harnessed for biomass and bioenergy applications.

Keywords River Ganges · River Yamuna · Confluence (*Sangam*) · CAZymes · Lignocellulosic

Introduction

Industrial production of several value-added products mainly counts on fossil fuels. However, the ever-increasing energy requisite and a dearth in the availability of fossil fuels have challenged us in search of alternate energy sources (Singhvi and Kim 2020). In recent years, lignocellulosic biomass (LCB) has emerged as a promising, cost-efficient, and most abundantly available, renewable bioresource to produce several value-added products (Serrano-Ruiz et al. 2011). Lignocellulose is mainly composed of cellulose (40–50%), hemicelluloses (25–35%), lignin (15–20%), while the other components such as pectin's, glycosylated proteins, and several other inorganic materials, that contribute a dismal fraction (Ma et al. 2020). The intricate networking of non-covalent and covalent cross-linkages in lignocellulose is the primary bottleneck that hinders the valorization of LCB into value-added products (Singhvi and Gokhale 2019; Sukumaran et al. 2021). Biodegradation of LCB is advantageous over physical and chemical methods in being an eco-friendly and a more proficient approach (Singhvi and Kim 2020; Bhatia et al. 2020). Microbial degradation and bioconversion of LCB into commercially important bio-ethanol and other platform chemicals are now being recognized globally (Doi and Kosugi 2004; Rittmann 2008; Carroll and Somerville 2009; Singhvi and Gokhale 2013; Van Meerbeek et al. 2019; Kameshwar et al. 2019).

The extrinsic lignocellulolytic potentials of certain culturable bacteria and fungi with an essential portfolio of metabolic enzymes called carbohydrate-active enzymes (CAZymes) has been extensively investigated in the past

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25 years (Kameshwar et al. 2019; Barrett et al. 2020) These enzymes center the anabolism and catabolism of complex carbohydrates, and essentially comprise of glycosyltransferases (for assembly), glycoside hydrolases, polysaccharide lyases, carbohydrate esterases, auxiliary activity enzymes and carbohydrate-binding modules (for breakdown) (Cantarel et al. 2009). In the last decade, metagenomics and next-generation sequencing (NGS) have facilitated deep insights into microbial genomes for understanding the functional potentials of unculturable microbes (Ndeh et al. 2017; Chuzel et al. 2018; Bertucci et al. 2019; Garron and Henrissat 2019; Gong et al. 2020). Therefore, functional annotation of a metagenomic dataset will allow us a comprehensive perception of different types of CAZymes, providing baseline information required to conduct further research (Montella et al. 2017).

River confluences are a dynamic ecosystem that harbors unique microbial and functional diversity (Samson et al. 2019, 2020). River Ganges and River Yamuna are two sacred rivers of India, originating from the mountain of the Himalayas and converging at *Prayagraj*. The confluence (commonly known as *Sangam*) of these two rivers at *Prayagraj* is highly revered, where *Kumbh Mela*, the world's largest bathing event, takes place. Despite their irrevocable significance, these rivers are in dismal conditions due to accelerating industrialization and anthropogenic activities. Both of these rivers originate from pristine sites; however, they are exposed to a range of pollutants before joining each other during their course of flow.

In the past few decades, terrestrial-lotic linkages and human interventions have enhanced the accumulation of carbon-rich substances along the River Ganges (Reddy et al. 2019). Direct discharge of effluent and waste from grossly polluting units, especially paper and pulp mills in River Ganges along with Kanpur, *Prayagraj*, and Varanasi, is of alarming concern (CPCB (Central Pollution Control Board 2013; Choudhury et al. 2018). Given that these rivers receive copious amounts of carbohydrate-rich wastes, analysis of CAZyme gene abundance therein could help understand the potentials of microbial diversity in the breakdown of these wastes. In view of this, we hypothesized the presence of a wide range of CAZymes in the unique ecological niche of the confluence of River Ganges and River Yamuna. This study also offers a broader scope for further exploration of the lignocellulolytic potentials from the microbes across river confluences, to address the challenges in biomass utilization and its conversion into value-added products.

Materials and methods

Sample site description and sample collection

The metagenomic datasets generated from sediment samples across the confluence stretch of River Ganges and River Yamuna in December 2017 are used in this study (Samson et al. 2019, 2020). An approximate of 340 kms transect of River Ganges across its pollution ascent was explored for CAZyme diversity. A total of 21 sediment samples (depth = 20 cm) were collected from 7 locations in triplicates (G14DS_Kanpur downstream, G15US_Prayagraj upstream; G15Y_Prayagraj Yamuna; G15S_Prayagraj *Triveni Sangam*; G16US_ Varanasi upstream, and G16DS_ Varanasi downstream) (Figure S1). For each location, the samples were collected from the left, middle, and right banks of the river and pooled together to obtain total community DNA. Further, the extracted metagenomic DNA were subjected to library preparation using long read nanopore sequencing platform of MinION. The library preparation was carried out using native barcoding kit (NBD103) and Ligation sequencing kit (SQK-LSK108), and loaded onto FLO-MIN-108D and run for 48 h.

In-silico analysis of the metagenomic datasets for CAZyme diversity

The raw reads obtained after MinION nanopore sequencing were subjected to quality check and statistical summary of reads were performed using NanoPack (De Coster et al. 2018). Functional annotation of Carbohydrate active enzymes (CAZyme) was performed by aligning the metagenomic reads to the GeneBank carbohydrate-active enzyme database (Zhang et al. 2018) using DIAMOND (–e lvalue 1e-10, –maxtargetseq 1) alignment tool (Buchfink et al. 2014). The data has been submitted in NCBI Sequence Read Archive (SRA) repository.

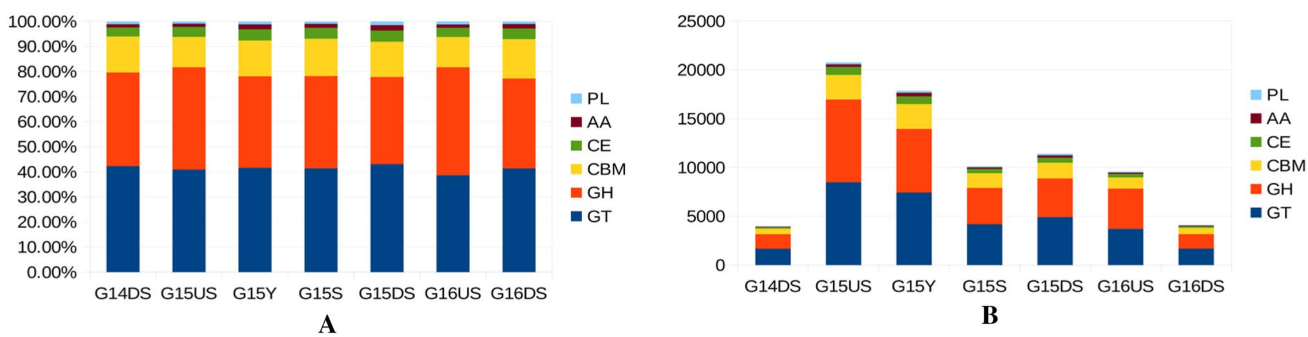
The Principal Coordinates Analysis (PCoA) of CAZyme genes data was constructed for beta diversity estimation using counts-based Bray–Curtis dissimilarity measure using microbiome package in R (Lahti et al. 2014). Data normalization was performed using qiime scipt *single_rarefaction.py* (Caporaso et al. 2010). The data distribution was checked using the Shapiro–Wilk normality test (p-value threshold of 0.05), followed by the construction of the Q-Q plot. The dissimilarity significance was assessed by anova test (p-value threshold of 0.05). Reads mapped to CAZyme database were extracted to investigate the microbial flora contributing to the carbohydrate active enzymes (CAZyme). Kaiju program was used for further taxonomic classification of reads (Menzel et al. 2016). Gephi, open-source software, was used for the visualization of data through dynamic network plots.

Results and discussion

The sediment samples were collected at a depth of 20 cm form left bank, right bank and from the centre of the river for each location (i.e. $n = 3$ per sampling site). Further processing from these samples involves pooling of the sediment samples from each sampling site followed by the metagenomic DNA extraction. Besides several studies have sampled sediments from 0 to 30 cm in depth to encompass total microbial diversity and their functions ((Sadaiappan et al. 2019; Zhao et al. 2020; Garber et al. 2021) Shotgun metagenome sequencing of the DNA extracted from the sediment samples generated 1.82 million reads, yielding around 7.9 Gb bases with an average quality score and sequence length of 10.2 and 4.5 kb, respectively (Table S1).

Distribution of carbohydrate-active enzymes in the metagenomes

River confluences are composed of a highly diverse and complex microbiome encoding a broad selection of multifunctional CAZymes that serve as a promising reservoir for application in the conversion of plant biomass to value-added products. Sediment microbial communities within freshwater ecosystems support important functions due to their high abundance, diversity, and stability. The confluence of River Ganges and Yamuna is a unique confluence that experiences anthropogenic disturbances in terms of accumulation of carbon-rich substrates due to the discharge of wastewater from adjoining paper and pulp industries, which could potentially contribute to the natural enrichment of LCB utilizing genes (CPCB (Central Pollution Control Board 2013; Choudhury et al. 2018). In view of this we analyzed the metagenome datasets across the confluence of Ganges and Yamuna River for CAZyme diversity. Previous studies on CAZymes are restricted to the gut, rumen, agricultural wastes, and wastewaters (Al-Masaudi et al. 2017; Wang et al. 2019; Jin et al. 2019; Ma et al. 2020).



	GT	GH	CBM	CE	AA	PL
G14DS	1680	1485	571	143	52	42
G15US	8490	8486	2517	828	265	184
G15Y	7436	6519	2556	789	361	204
G15S	4187	3723	1504	441	163	93
G15DS	4913	3972	1603	510	245	166
G16US	3706	4132	1153	354	126	115
G16DS	1696	1472	644	175	74	40

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Figure. 1. **A** Relative percentage of CAZyme Class distributed at each location. **B** Relative abundance of CAZyme Class distribution at each location. **C** An overview of CAZymes identified in the metagenomes

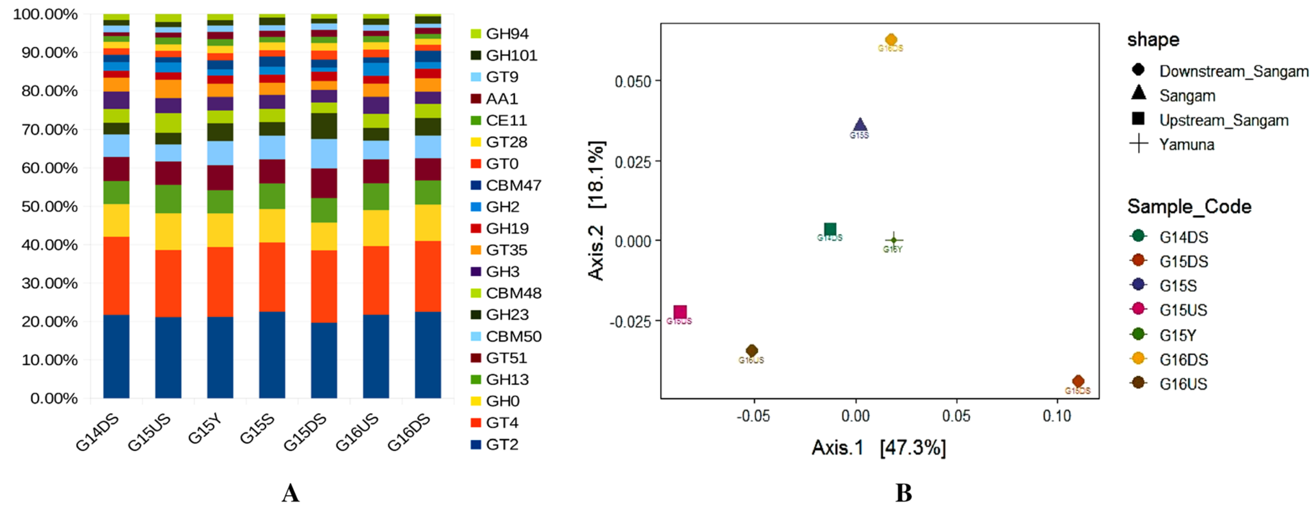

Figure. 2. **A** Relative percentage of top 20 CAZyme families distributed at each location. **B** Principal Coordinate Analysis (PCoA) estimation of the samples with CAZyme families

However, there are very limited reports on CAZyme diversity in freshwater ecosystems (Andrade et al. 2017a; Xie et al. 2020). To the best of our knowledge, this is the first report to provide insights into the CAZyme diversity across the sacred confluence of River Ganges and River Yamuna. The metagenome analysis revealed the presence of six different categories (CBMs: carbohydrate-binding modules; CEs: carbohydrate esterases; GHs: glycoside hydrolases; GTs: glycosyltransferases; PLs: polysaccharide lyases and AAs: auxiliary activities) of CAZymes at all the locations (Fig. 1A). GT and GH were the most abundant CAZyme families at all the locations, with a relative percentage of (41–43%) and (36–40%) respectively (Fig. 1B). Enzymes from GT families are responsible for synthesis of glycoside linkages by transfer of a sugar moiety from a donor substrate to an acceptor, while those from GH are involved in the metabolism of substrates like starch, cellulose, chitin, and xylan. (Lairson et al. 2008; Stewart et al. 2018). CBMs are a large group of protein domains that increase the proximity of the GH enzymes to its substrate, especially for insoluble substrates, and thereby enhance the enzyme hydrolysis (Sidar et al. 2020). Overall, the functional annotation to understand the gene diversity of microbes for carbohydrate metabolism revealed a total of 287 CAZyme families (Fig. 2A, Fig. 3). The most abundant CAZyme families detected were GT2 (21–22%) and GT4 (17–20%). The enzymes belonging to these families are known to be mainly involved in the biosynthesis of cellulose, chitin, and sucrose (Pinard et al. 2015). Whereas GH0 (8–9%), GH13 (6–7%), GT51 (5–6%), CBM50 (4–7%), GH23 (3–6%), CBM48 (2–5%), GH3 (3–4%), GT35 (2–4%), GH19 (1–2%), GH2 (1–3%), CBM47 (1–2%), GT0 (1–2%), GT28 (1–2%), CE11 (1.2–1.8%), AA1 (0.9–1.8%), GT9 (1–1.7%), GH101 (1.2–2%), GH94 (0.5–2%) respectively, are among the top 20 CAZyme families. These enzymes and enzyme moieties are responsible for the degradation of complex carbohydrates like lignocellulose and chitin (Chettri et al. 2020). The presence of a plethora of CAZy genes across the confluence stretch of River Ganges and Yamuna is indicative of the potential of various microbes' utilizing carbohydrates as their main substrates. This can be supported by a recent study on the River Ganges at Varanasi, where analysis of the functional profile from microbes revealed dominant metabolic pathways to be associated with carbohydrate metabolism (Reddy et al. 2019). Further the beta diversity analysis indicated that, as compared to other locations across the confluence stretch, G15DS appears to be more diverse in CAZyme composition (Fig. 2B). This could be correlated with our previous study wherein, the beta diversity analysis showed G15 DS as a highly diverse site in terms of microbial diversity (Samson et al. 2019). Therefore, the downstream of confluence could probably provide a better environmental niche for the exploration of microbes with diverse set of CAZymes, which could be utilized further for conversion or degradation of lignocellulosic biomass.

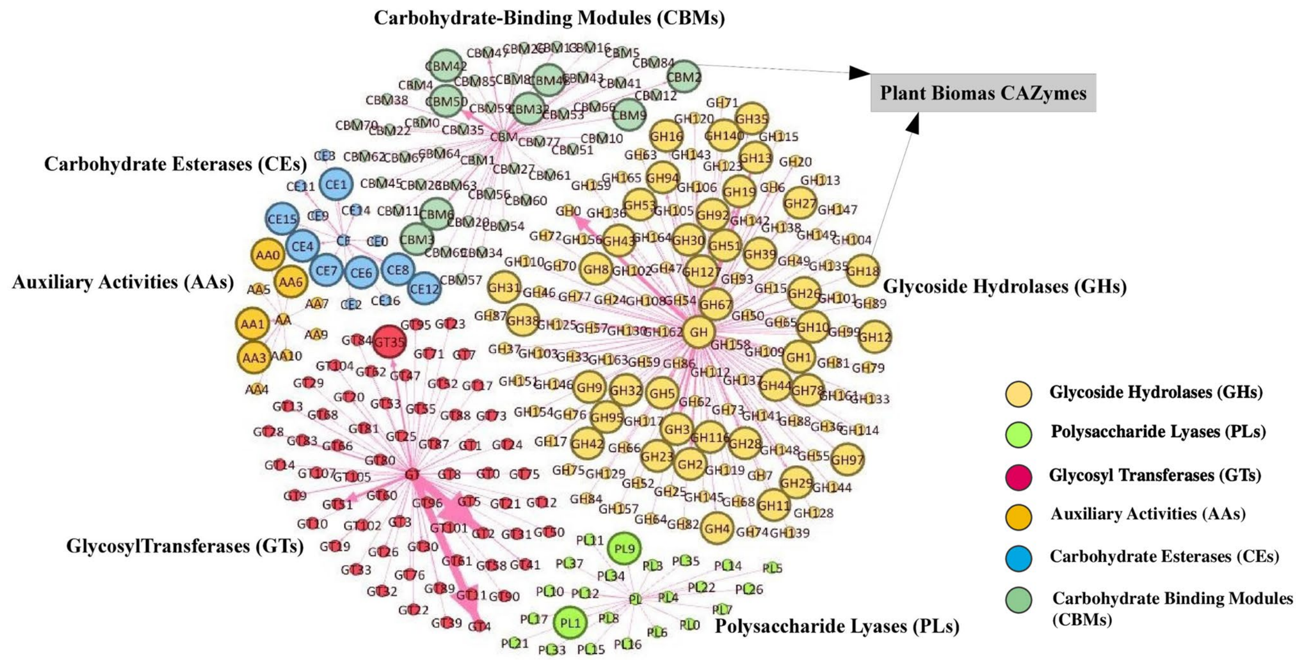

Fig. 3 A nexus of overall CAZyme genes from each CAZyme class and those involved in plant biomass degradation (large nodes)

Plant-biomass degrading CAZymes

It is a well-known fact that River Yamuna is the longest and most contaminated tributary of River Ganges untreated or partially treated domestic sewage, industrial effluents and agricultural effluents are the major contributors of carbohydrate rich pollutants in the river (Dwivedi et al. 2020). In view of this we further analyzed the metagenomic datasets for presence of microbes with LCB degrading enzymes. The analysis revealed the presence of plant biomass hydrolyzing cellulases from GH3, GH5, GH8, GH9, GH12, GH44, GH116, GH140 families. In our study, the predominant oligosaccharide-degrading enzyme was the GH3 family, which encodes β-glucosidase, xylanase, β-1,4-xylosidase, endoglucanases, cellobiohydrolases, cellodextrinases and glucosylceramidase. These enzymes play a crucial role in the cellulose degradation process (Escuder-Rodríguez et al. 2018). Additionally, genes responsible for the degradation of hemicellulose include enzymes from GH1-GH5, GH10, GH11, GH26, GH27, GH30, GH31, GH 38, GH39, GH43, GH53, and GH92 (Fig. 3). These results corroborate with several metagenomic studies focusing on plant fiber degradation (Comtet-Marre et al. 2017; Wang et al. 2019; Ma et al. 2020). Besides, the GH family, we also observed the presence of genes from the CBM family, with CBM 48 and CBM 50 being the most abundant. Details of the gene families involved in plant biomass hydrolysis have been tabulated (Table 1).

Microbial community analysis for CAZyme diversity and LCB hydrolysis

Analysis of the metagenomic datasets for microbes with putative CAZyme reads across the confluence stretch revealed a predominance of bacteria (97%) followed by archaea (1.67%), viruses (0.7%), and fungi (0.5%). The major bacterial phyla in this study with presence of CAZymes include *Proteobacteria* (45%)*, Bacteroidetes* (11%)*, Chloroflexi* (9%)*, Acidobacteria* (8%), *Verrucomicrobia* (7%)*, Planctomycetes* (6%)*, Actinobacteria* (4%)*, Nitrospirae* (3%), and *Firmicutes* (2%) (Fig. 4A, Fig. S3). *Proteobacteria* was to be the most dominant phylum, comprising a significant percentage of CAZymes and proteins associated with lignocellulose degradation. This is in contrast with other studies wherein, *Bacteroidetes* phylum is reported to be the most abundant and important microbial component

Table 1 An Overview of plant biomass processing enzymes identified in the metagenomes

		G14DS	G15US	G15Y	G15S	G15DS	G16US	G16DS
Cellulose degradation								
GH5	Cellulase, endoglucanase, betamannosidase, Mannan endo-1,4-beta-mannosidase	28	110	94	57	55	67	18
GH8	endoglucanase	2	16	13	4	8	7	1
GH9	endoglucanase	14	50	38	18	18	21	13
GH12	Cellulase, endoglucanase	0	1	2	3	0	0	0
GH140	Endoglucanase	7	33	31	7	12	17	3
GH44	Endoglucanase, beta-mannanase	4	16	10	8	4	4	2
GH116	Beta-glucosidase, Glucosylceramidase	3	26	11	11	6	26	1
GH3	Glucan 1,4-beta-glucosidase, beta-glucosidase, beta-*N*-acetylhexosaminidase, betalactamase	108	515	391	219	232	245	79
Hemicellulose degradation								
GH10	Endo-beta-1,4-xylanase, xylanase	5	21	24	13	14	14	4
GH11	Endo-beta-1,4-xylanase	0	0	1	1	2	1	0
GH1	Beta-glucosidase, glycosidase	32	156	98	62	63	57	23
GH42	beta-glucosidase	5	36	35	16	15	21	4
GH2	Beta-mannosidase, beta-galactosidase, beta-glucuronidase, beta-glucosidase	54	340	180	128	80	190	44
GH26	Mannan endo-1,4-beta-mannosidase, Beta-mannanase	3	15	8	8	3	8	0
GH27	Alpha-galactosidase	4	13	12	4	3	13	3
GH3	xylosidase/arabinosidase, beta-*N*-acetylhexosaminidase, beta-xylosidase, Beta-lactamase	108	515	391	219	232	245	79
GH31	Alpha-xylosidase, alpha-glucosidase	22	128	75	55	42	80	18
GH38	Alpha-mannosidase, Mannosylglucerate hydrolase	17	93	53	31	27	50	18
GH39	Beta-xylosidase, *O*-antigen polymerase	4	41	23	15	9	17	4
GH4	Alpha-galactosidase, 6-phospho-beta-glucosidase	22	94	59	43	15	49	21
GH43	Beta-xylosidase, arabinoxylan arabinofuranohydrolase, Dipeptidyl aminopeptidase/acylaminoacyl-peptidase, Arabinan endo-1,5-alpha-L-arabinosidase	34	191	136	97	78	100	27
GH5	Beta-mannosidase, mannanase, endoglucanase	28	110	94	57	55	67	18
GH53	Arabinogalactan endo-1,4-beta-galactosidase	0	3	3	2	2	6	0
GH92	Alpha-1,2-mannosidase	12	51	35	27	15	51	14
GH30	glucuronoxylanase, *O*-glycosyl hydrolase, Glucosylceramidase	12	45	35	26	12	27	5
CE1	Carbohydrate acetyl esterase, 1,4-beta-xylanase	27	89	76	36	31	36	19
CE4	Xylanase/chitin deacetylase, polysaccharide deacetylase	21	116	123	63	118	44	22
CE7	Acetyl xylan esterase	2	39	25	19	12	14	7
CE15	Acetyl xylan esterase	6	51	42	24	16	17	8
CE6	Acetyl xylan esterase	1	8	13	5	7	3	6
GH78	Alpha-rhamnosidase	21	104	72	39	23	74	8
GH28	Endopolygalacturonase (pectin depolymerase)	23	109	109	66	67	54	23
GH35	Beta-galactosidase, mannonate dehydratase	6	40	28	11	19	20	7
PL1	Pectate lyase	6	24	35	14	17	16	4
PL9	Pectate lyase	3	17	22	7	12	6	4
CE8	Pectin methylesterase, Pectinesterase, polysaccharide lyase	13	54	81	42	63	29	17
CE12	Pectate lyase, rhamnogalacturonan acetylesterase	1	26	26	23	10	17	10
GH18	Chitinase	5	23	32	15	19	21	4
GH19	Chitinase, carboxypeptidase	44	254	241	131	177	114	64
GH3	Beta-*N*-acetylhexosaminidase, beta-xylosidase, Beta-lactamase, xylosidase/arabinosidase	108	515	391	219	232	245	79

Table 1 (continued)

		G14DS	G15US	G15Y	G15S	G15DS	G16US	G16DS
CE4	Chitin deacetylase/xylanase, polysaccharide deacetylase	21	116	123	63	118	44	22
GH13	Alpha amylase, malto-oligosyltrehalose trehalohydrolase, alpha glucosidase, glycogen debranching enzyme GlgX, Glycosidases,glucan 1,4-alpha-maltotetraohydrolase,alpha-1,6-glucosidase, pullulanase-type, trehalose synthase, sucrose phosphorylase, alpha-D-1,4-glucosidase, Maltodextrin glucosidase, amylosucrase	146	970	678	413	463	393	161
GH23	Lytic transglycosylase catalytic, murein transglycosylase, Peptidoglycan-binding LysM	74	400	523	219	490	187	118
GH51	Alpha-*N*-arabinofuranosidase, exo-alpha-L-arabinofuranosidase 2	22	113	61	34	25	45	10
GH67	Alpha-glucuronidase	3	9	12	7	2	4	1
GH127	Non-reducing end beta-L-arabinofuranosidase	10	76	37	33	13	43	8
GH32	Beta-fructofuranosidase, raffinose invertase, glycosidase	7	25	27	14	11	25	7
GH97	Alpha-glucosidase	16	71	42	27	28	39	6
GH16	Glucan endo-1,3-beta-D-glucosidase, laminarinase	10	37	51	40	34	39	13
GH29	Alpha-L-fucosidase	27	201	111	65	30	116	23
GH95	Alpha-L-fucosidase	18	121	67	31	23	79	11
GH94	Cellobiose phosphorylase, glycosyltransferase 36	38	272	179	57	87	67	15
GT35	Starch phosphorylases	88	614	383	196	167	193	89
CBM3	Cellulose Binding module	6	23	18	17	9	2	8
CBM32	Cellulose Binding module	29	123	111	66	54	57	36
CBM48	Glycogen debranching enzymes like GH13 family	87	672	374	214	200	206	94
CBM50	Chitin or peptidoglycan binding	143	581	706	383	556	275	151
CBM6	Hemi-cellulose binding module	24	83	97	61	68	60	27
CBM9	Cellulose binding module	13	51	61	48	44	39	19
AA1	Multicopper oxidase	24	168	212	101	130	78	41
AA3	Choline dehydrogenase	21	68	114	48	93	34	26
AA6	Quinone oxidoreductase	0	3	3	1	1	1	0

for the breakdown of polysaccharide wastes (Wang et al. 2019; Puentes-Téllez and Salles 2020). Certain members of *Proteobacteria*, *Bacteroidetes, Actinobacteria,* from a wide range of ecological niches, contain 'Polysaccharide Utilization Loci' (PULs). These PULs are discrete gene clusters of CAZymes and the associated transporters, that collectively enable the metabolism of carbohydrate-rich substrates (Ausland et al. 2021). Further analysis of the major bacterial phyla for the presence of plant biomass-degrading CAZymes revealed most enzymes from GH category (Fig. 4B).

Besides bacteria, several candidate archaeal phyla from the metagenomic datasets also showed presence of CAZyme sequences. Interestingly, we observed archaea from *Thaumarchaeota, Euryarchaeota, Candidatus Bathyarchaeota; Candidatus Lokiarchaeota, Crenarchaeota, and Candidatus Thermoplasmatota* (Figure S4). Further, the archaeal diversity analysis revealed presence of CAZyme contigs from GH families in several candidate archaeal phyla. There are very limited reports on archaeal CAZymes from the freshwater ecosystems such as rivers (Andrade et al. 2017b; Wang et al. 2020; Tran et al. 2021). GH from archaea is characterized as hyperthermozymes with remarkable thermostability and thus serves as a potential candidate for high thermal industrial processes involving efficient biomass conversion and biofuel generation (Helbert et al. 2019; Strazzulli et al. 2020; Chettri et al. 2020). This finding opens up new horizons for exploration of the confluence stretch of River Ganges and River Yamuna for hyperthermozymes, which can then be used as a sustainable biological solution for efficient thermo-active and thermo-stable transformation of recalcitrant plant biomass.

In conclusion, our study is a first ever report that highlighted presence of a vast and diverse reservoir of CAZyme sequences associated with the microbial communities across the confluence stretch of River Ganges and Yamuna. Metagenomic analysis and functional annotation of the microbial community with putative CAZyme reads discovered microbial populations with the potentials to

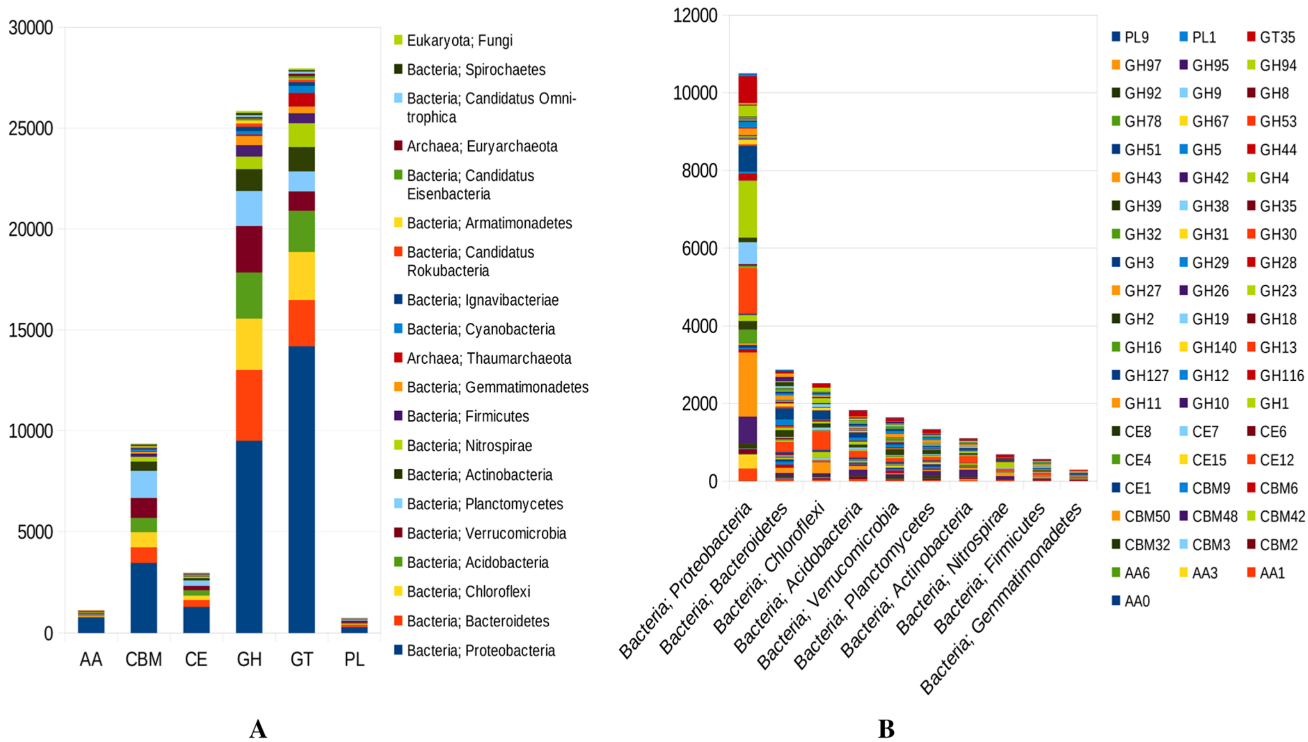

Figure. 4. **A** Relative abundance of top 20 microbial community contributing to CAZyme classes across the confluence stretch. **B** Microbial community analysis of plant biomass degrading CAZymes across the confluence stretch

depolymerize and degrade complex lignocellulosic biomass. The higher abundance of GH enzyme families potentially indicates the high levels of organic matter due to the increasing pollution. This work provides further avenues of opportunity of exploring the detailed biochemical and structural functions of each CAZyme from the microbes inhabiting therein and use the novel enzyme candidates for future biotechnological and industrial applications.

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Author contributions All the authors have read and review the manuscript for submission. VR: analysis and writing original draft; RS: writing, editing, sample collection and sequencing RKY: editing and sequencing; MD: supervision, designing study, and reviewing.

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Availability of data and materials The raw sequencing data have been deposited at NCBI under the project accession number PRJNA716890. The samples raw data were deposited under NCBI SRA accession number SRX10447184, SRX10447183, SRX10447182, SRX10447181, SRX10447180, SRX10447179, SRX10447178, respectively.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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